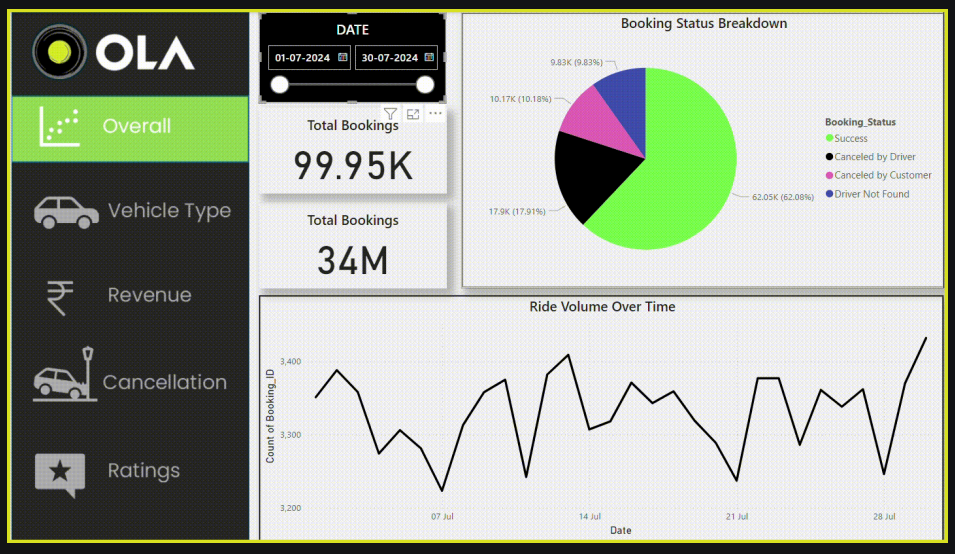
PORTFOLIO PROJECT REPORT

Ride Booking Analysis

SQL and Power BI Project

A portfolio project exploring booking performance, ride trends, cancellations, vehicle types, and customer experience.

Tools: **MySQL** • **SQL** • **Power BI**



Synthetic dataset • Educational analysis

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1. Project Overview

This project analyses ride-booking data for an OLA-style mobility service in Bengaluru. SQL is used to query booking records and answer operational questions, while Power BI is used to present key metrics and trends in an interactive dashboard.

> **Dataset note** The project materials describe a generated/synthetic dataset. The results should therefore be presented as an educational analysis, not as official OLA company data or verified business performance.

2. Project Objectives

- Review booking status and successful-booking patterns.
- Compare ride distance and customer ratings across vehicle types.
- Count customer and driver cancellations and review recorded reasons.
- Identify customers with the highest number of bookings.
- Calculate booking value for successful rides.
- Visualise booking volume over time and booking-status distribution in Power BI.

3. Tools and Technologies

TOOL	PURPOSE
MySQL	Store and query booking data.
SQL	Filter, aggregate, rank, and create reusable views.
Power BI	Build KPI cards, date filtering, and charts.

4. Dataset Description

The analysis uses one month of synthetic ride-booking data for Bengaluru, stored in `Bookings.csv`. The dataset covers booking details, ride characteristics, payment information, cancellation information, and customer and driver ratings.

FIELD(S)	DESCRIPTION
BOOKING DETAILS	
Date and Time	Booking date and time
Booking_ID	Unique booking identifier
Booking_Status	Outcome/status of the booking
Customer_ID	Customer identifier
Vehicle_Type	Vehicle category
RIDE CHARACTERISTICS	
Pickup_Location / Drop_Location	Trip origin and destination
V_TAT / C_TAT	Vehicle-arrival and customer-arrival time measures
Ride_Distance	Trip distance
CANCELLATIONS AND INCOMPLETE RIDES	
Canceled_Rides_by_Customer / reason	Customer cancellation details
Canceled_Rides_by_Driver / reason	Driver cancellation details
Incomplete_Rides / Incomplete_Rides_Reason	Incomplete trip indicator and reason
PAYMENT AND RATINGS	
Booking_Value / Payment_Method	Booking amount and payment type
Driver_Ratings / Customer_Rating	Ratings associated with a trip

5. SQL Analysis

The queries below answer the core questions covered in the project and run against the `bookings` table imported from `Bookings.csv`. Text filters, such as cancellation status values, match the values stored in the data.

5.1 Retrieve successful bookings

```sql
SELECT *
FROM bookings
WHERE Booking_Status = 'Success';
```

5.2 Average ride distance by vehicle type

```sql
SELECT
    Vehicle_Type,
    AVG(Ride_Distance) AS avg_ride_distance
FROM bookings
GROUP BY Vehicle_Type;
```

5.3 Count customer-cancelled bookings

```sql
SELECT COUNT(*) AS customer_cancelled_bookings
FROM bookings
WHERE Booking_Status = 'Canceled by Customer';
```

5.4 Top five customers by number of bookings

```sql
SELECT
    Customer_ID,
    COUNT(Booking_ID) AS total_rides
FROM bookings
GROUP BY Customer_ID
ORDER BY total_rides DESC
LIMIT 5;
```

5.5 Driver cancellations due to personal or car-related issues

```sql
SELECT COUNT(*) AS driver_cancellations_personal_car
FROM bookings
WHERE Canceled_Rides_by_Driver = 'Personal & Car related issue';
```

5.6 Maximum and minimum driver ratings for Prime Sedan

```sql
SELECT
    MAX(Driver_Ratings) AS max_driver_rating,
    MIN(Driver_Ratings) AS min_driver_rating
FROM bookings
WHERE Vehicle_Type = 'Prime Sedan';
```

5.7 Retrieve bookings paid using UPI

```sql
SELECT *
FROM bookings
WHERE Payment_Method = 'UPI';
```

5.8 Average customer rating by vehicle type

```sql
SELECT
    Vehicle_Type,
    ROUND(AVG(Customer_Rating), 2) AS avg_customer_rating
FROM bookings
GROUP BY Vehicle_Type;
```

5.9 Total booking value for successful rides

```sql
SELECT
    SUM(Booking_Value) AS total_successful_booking_value
FROM bookings
WHERE Booking_Status = 'Success';
```

5.10 Incomplete rides and their reasons

```sql
SELECT
    Booking_ID,
    Incomplete_Rides_Reason
FROM bookings
WHERE Incomplete_Rides = 'Yes';
```

6. Power BI Dashboard

The dashboard includes a date-range slicer, KPI cards, a booking-status breakdown pie chart, and a ride-volume-over-time line chart. These visuals provide a starting point for exploring booking outcomes and daily booking patterns.

- **Date-range slicer:** filter the displayed results to a selected period.
- **Booking Status Breakdown:** compare the share and count of booking outcomes.
- **Ride Volume Over Time:** inspect daily changes in booking volume.
- **KPI cards:** display clearly named metrics such as total bookings and total booking value.

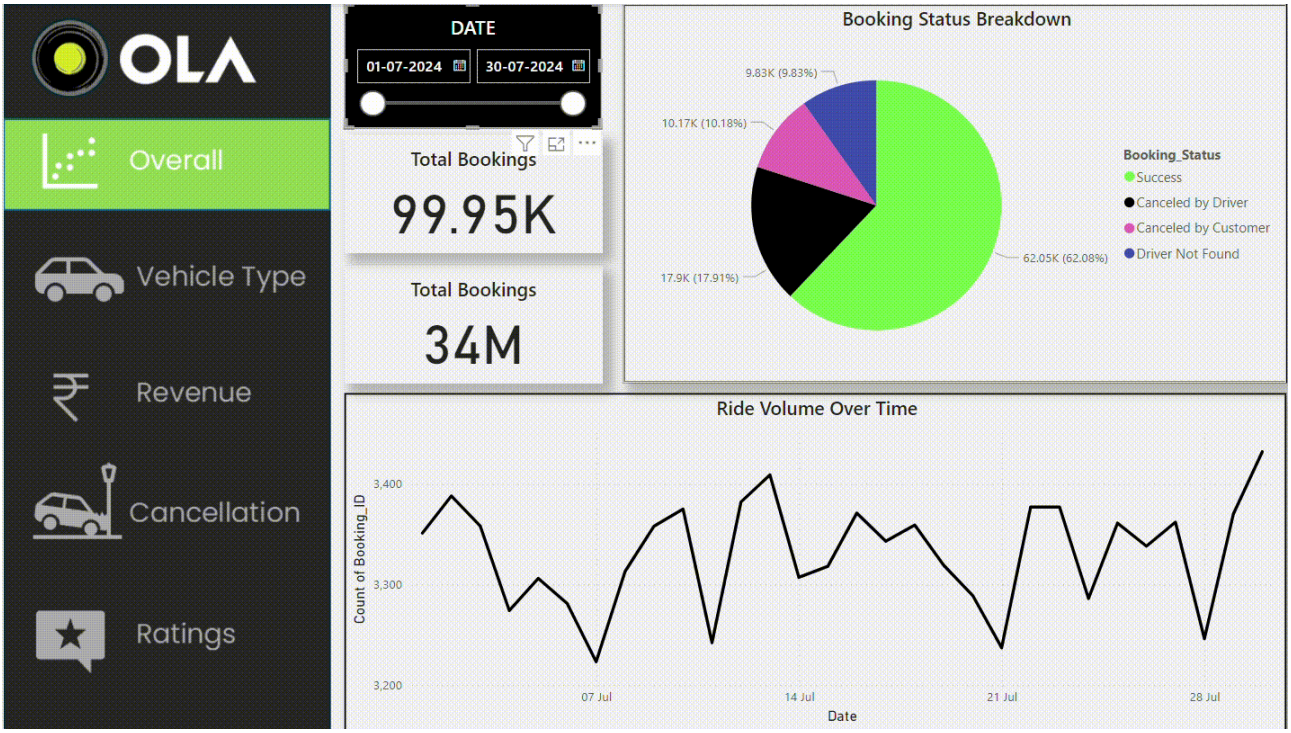


Figure 1. Overall page of the Ride Booking Analysis dashboard (1–30 July 2024)

7. Business Insights

Key findings from the Power BI dashboard for 1–30 July 2024.

01

Most bookings succeed, but over a quarter are cancelled

SOURCE: DASHBOARD · OVERALL AND CANCELLATION PAGES

62.08% of 99.95K bookings succeeded (62.05K); **28.09%** were cancelled (28.07K) and 9.83% ended with no driver found.

Driver cancellations (17.91%) were higher than customer cancellations (10.18%).

02

Cancellation reasons

SOURCE: DASHBOARD · CANCELLATION PAGE

Customer cancellations: “Driver is not moving towards pickup” (30.3%) and “Driver asked to cancel” (25.39%) account for over half.

Driver cancellations: “Personal & Car related issue” is the leading reason (35.48%), followed by “Customer related issue” (29.34%).

03

Auto rides are much shorter than other vehicle types

SOURCE: DASHBOARD · VEHICLE TYPE PAGE

Average distance travelled is 10.04 for Auto, against 24.91–25.14 for the other six vehicle types.

Prime Sedan has the highest total booking value (8.06M); the others range from 7.68M to 7.93M.

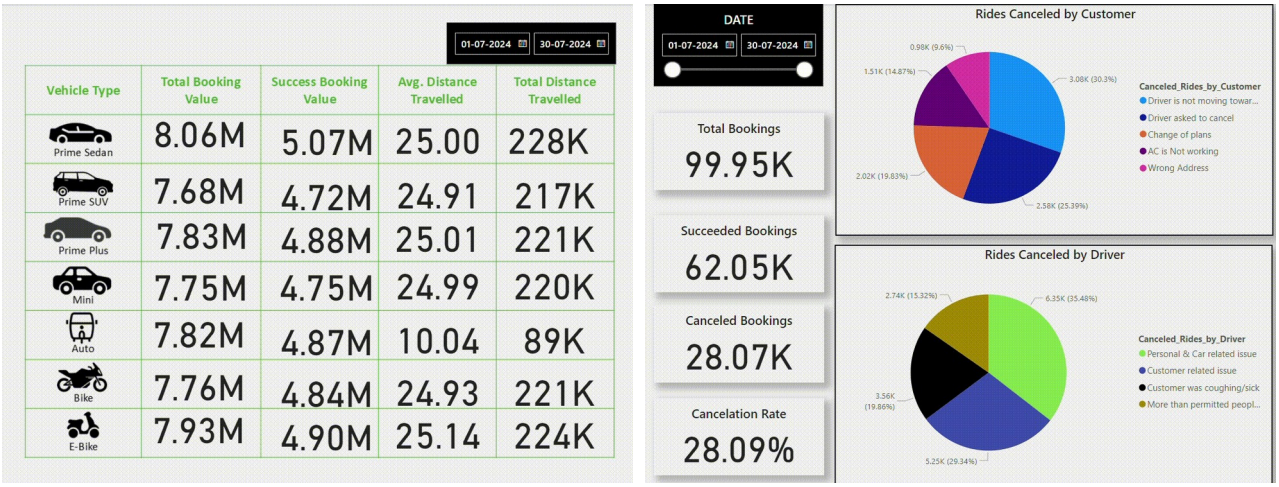


Figure 2. Vehicle Type (left) and Cancellation (right) pages of the dashboard

8. Business Questions This Project Supports

- What proportion of bookings are successful or cancelled?
- How does ride distance differ by vehicle type?
- Which customers have the highest booking counts?
- How many cancellations are associated with selected reasons?
- What is the total booking value of successful rides?
- How does booking volume change over the selected dates?
- How do customer ratings vary across vehicle types?

9. Key Learnings

- Using `WHERE` to filter records based on business conditions.
- Using `COUNT`, `SUM`, `AVG`, `MIN`, and `MAX` to calculate summary metrics.
- Grouping data with `GROUP BY` and ranking results with `ORDER BY` and `LIMIT`.
- Creating SQL views to save reusable query logic.
- Presenting metrics and time-based trends with Power BI visuals.

10. Project Resources

Dataset	Bookings.csv
SQL script	Ola DA Project SQL.sql
Dashboard	Power BI dashboard (see Section 6)
Repository	https://github.com/Magu2801/ola-ride-booking-analysis